

15.075 Exam 2

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Grading is based on demonstration of conceptual understanding, so you need to show all of your work.

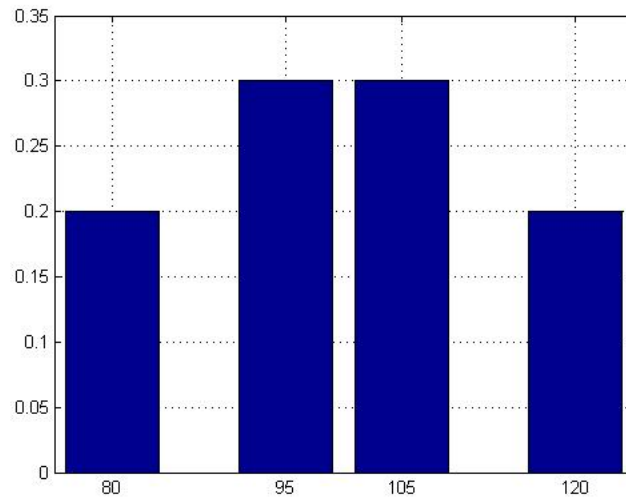
Problem 1

You are in charge of a study that compares how two weight-loss techniques (Diet and Exercise) affect the weight loss of overweight patients. The table below shows the number of obese and severely

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Problem 2

You are the owner of a bakery selling cheesecakes. Some days you sell a lot of cheesecakes and some other days you don't, independently of other days. In fact, the distribution of the number of cheesecakes you sell each day is shown below.



If we measure cheesecakes sales for 100 days, what is the approximate probability that the average cheesecakes sales (over the 100 days) will be more than 102 cheesecakes?

Problem 3

Suppose that the observations $X_1, X_2, \dots, X_n$ are iid with common mean θ and known variance σ^2 . Consider the following estimator of the mean θ :

$$\hat{\theta}_n = \frac{X_1 + \dots + X_n}{n+1}.$$

1. What is the bias of this estimator? What happens to the bias as we increase the size of the sample?
2. What is the MSE of the estimator?

Hint: Remember to use either what you know about $\text{Var}(X)$, or what you know about the variance of the sum of independent random variables.

Problem 4

The weight of an object is measured eight times using an electronic scale that reports the true weight plus a random error that is normally distributed with zero mean and variance $\sigma^2 = 4$. Assume that the errors in the observations are independent. The following results are obtained:

{4.45, 4.02, 5.51, 1.10, 2.62, 2.38, 5.94, 7.64}

Compute a 95% confidence interval for the true weight.

Problem 5

The returns of an asset management firm in different years are independent and normally distributed with unknown mean and variance. The asset management firm claims that the standard deviation of the returns is as low as $\sigma = 2\%$ and the mean of the returns is $\mu = 22\%$. You believe that this is too good to be true. To verify your suspicion, you take the returns from the last 10 years. These are:

$$r = \{20.6, 19.2, 17, 19.1, 18.7, 22.5, 27.2, 17.9, 22.5, 21.3\}$$

(To help you with the calculations, we give you that: $\sum_{i=1}^n r(i) = 206$ and $\sum_{i=1}^n (r(i) - \bar{r})^2 = 79.34$)

1. Calculate the sample mean and standard deviation of the above random sample.
2. What is the probability that when we draw a new random sample of size 10, its sample mean will be below the one you calculated in part 1? Assume that the company's claims are correct.
3. What is the probability that when we draw a new random sample of size 10, its sample standard deviation will be larger than the one you calculated in part 1, assuming that the company's claims are correct?

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